

Result Artifacts Summary

Manuscript: TRA-SAE: A Low-Cost Retrieval-Augmented Expert Reasoning System for Mixed Physics-Logic Question Answering with a 4B Language Model

Mapping of every manuscript table/figure to its generating script and raw artifact.

Item	Reports	Script	Raw artifact
T1	Dataset split	processed_data builder	exact_train / exact_val
T3 / Fig2	cfg0..cfg5 main results	step1_rerun_cfg0_3.py	logs/qwen35_ablation_canonical*.json
T3(B1-6)/Fig3	best-of-3 baselines	step8_fair_baselines.py	logs/fair_baselines_*.json
Retr.tbl	baselines + retrieval	step8_fair_baselines.py	logs/fair_baselines_retrieval_*.json
T4	Wilson CI + McNemar	step4_stats_and_errors.py	logs/stats_results.json
T5	multi-seed cfg3	step2_multiseed_cfg3.py	logs/...seed1337/seed2024.json
T6	MMLU physics	step9_external_benchmark.py	logs/external_benchmark_*.json
FOLIO	FOLIO logic	step9_external_benchmark.py	logs/external_benchmark_logic_*.json
T7	tool-augmented	step13_tool_baselines.py	logs/tool_baselines_results_latest.json
T8	reward ablation	step5_reward_ablation.py	logs/reward_ablation_results_latest.json
T9	compute/cost	training logs	logs/ (runtime/cost)
T10/T11	error taxonomy	step12_error_reannotation.py	logs/...canonical_partB.json
Audit	leakage check	step10_leakage_check.py	console report